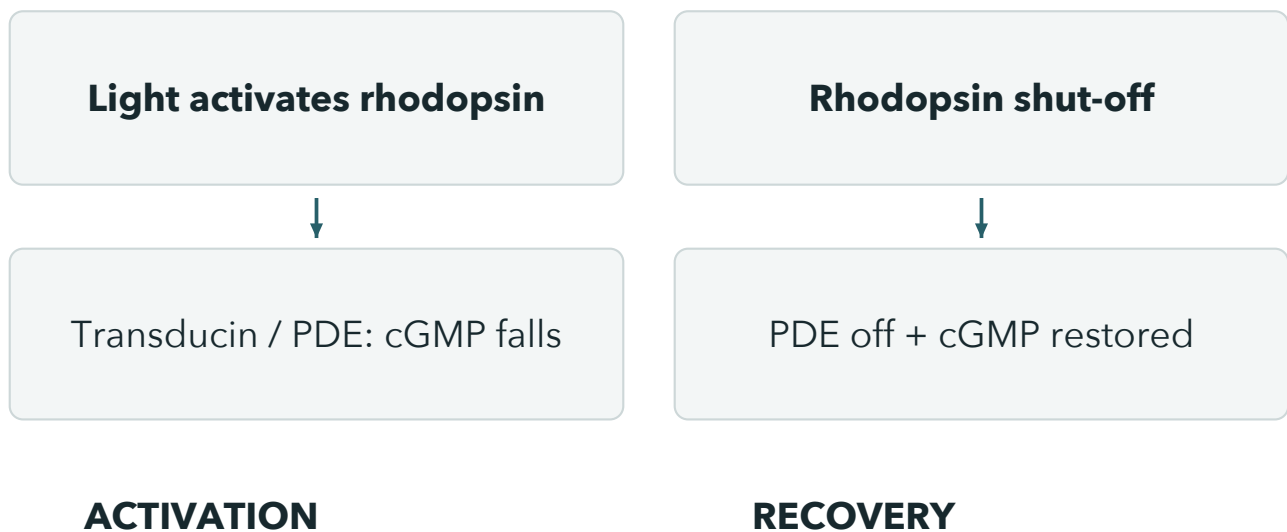


Phototransduction recovery and dark adaptation

Part 1 | Activation is only half the cascade

A rod must terminate signalling and rebuild cGMP before its channels reopen.



Initial response and termination can fail independently

Activation and recovery are separated so a prolonged response can be localised.

Read a time course

Normal onset with prolonged recovery suggests defective shut-off or cGMP restoration. An absent initial response suggests a different process. Signal deactivation and pigment regeneration are separate events; response shape constrains the hypothesis without identifying a unique protein.

Three recovery targets

Several molecular mechanisms terminate different active intermediates.

Target

Feature	Termination process	If delayed
Activated rhodopsin	Phosphorylation and arrestin binding	Continued transducin activation
Transducin-PDE activity	GTP hydrolysis, accelerated by RGS complex	Continued cGMP hydrolysis
Low cGMP	Guanylate-cyclase synthesis	Delayed channel reopening

Use rescue evidence

A protein-specific rescue is stronger than a nonspecific amplitude change.

Normal onset narrows the defect but does not uniquely identify it.

Keep the model explicit

Much mechanistic evidence comes from genetically modified animal rods. Apply the principle of active-intermediate shut-off without assuming identical kinetics in every species or cone subtype.

Calcium provides feedback

Channel closure lowers calcium entry and changes recovery regulation.

Feedback during illumination

- cGMP-gated channels close
- ↓
- Calcium entry falls while extrusion continues
- ↓
- GCAP-mediated cyclase regulation changes
- ↓
- cGMP restoration helps channels reopen

Interpretation

The feedback opposes an indefinitely sustained electrical response.

It helps adjust sensitivity and recovery as illumination changes.

The rate-limiting step depends on the preparation and response conditions.

Avoid a shortcut

A fall in calcium does not mean that every calcium-dependent process stops. Different proteins respond to calcium binding in different ways; identify the specific regulatory link in the proposed experiment.

Dark-adaptation curves are functional evidence

Early cone recovery and later rod sensitivity can be analysed separately.

Curve feature

Feature	Main interpretation	Limit
Early threshold improvement	Cone-mediated recovery	Depends on bleach and target
Rod-cone break	Rod threshold becomes lower	Not a fixed universal time
Later low threshold	Rod-mediated sensitivity	Not a gene diagnosis

Standardise before comparing

- Control bleaching exposure
↓
- Control target position and size
↓
- Control pupil and fixation
↓
- Compare recovery and final threshold

Do not confuse three tests

Electrical activation, behavioural sensitivity and pigment availability answer related but different questions.

Method

Feature	Primary measurement	Useful caution
Flash response	Electrical waveform	Recording conditions matter
Dark adaptometry	Detection threshold over time	Psychophysical cooperation matters
Molecular assay	Pigment or protein state	Does not directly measure vision

A delayed recovery result

What additional evidence is available?

